INCH POUND

MIL-C-83810 (84)

4 October 1991

MILITARY SPECIFICATION

CRANE, FLOOR, PORTABLE, 1,500 POUND CAPACITY

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers a 1,500 pound lifting capacity portable floor crane for maintenance support of the F-16 Air Combat Fighter Aircraft.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-T-704	Treatment and Painting of Material
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component Chemical Agent Resistant
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic, Hydrocarbon Base, Aircraft
MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/LZDTD, Robins AFB, GA 31098-5365, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3950

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STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for Use on Drawings, and on Specifications, Standards and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-970	Specification and Standard Order of Precedence, for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-808	Finishes, Protective and Codes, for Finishing
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Nontactical, Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MS 21236-2	Ring, Cargo, Tiedown (10,000), Type III
MS 25488	Connector Plug, Attachable External Electric Power, Aircraft, 28 Volt DC Operating Power
MS 35000-4	Battery, Storage, Lead-Acid Waterproof
MS 51336	Lunette - Coupler, Drawbar, Ring
MS 90347	Cable Assembly, External Electric Power, Aircraft, 28 Volt DC Operating Power

(Unless otherwise indicated, copies of federal and military specifications, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issues of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

D1.1	Structural Welding Code, Steel

(Application for copies should be addressed to American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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AMERICAN WELDING SOCIETY (AWS)

Welding Qualifications

(Application for copies should be addressed to American Welding Society, 2501 NW Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J534	Lubrication Fittings
J959	Lifting Crane, Wire Rope Strength Factors

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INCORPORATED

Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Incorporated, 3200 West Market Street, Akron, OH 44313.)

3. REQUIREMENTS

3.1 Design

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced specification shall be suitable in every respect for this type of equipment. Wood shall not be used for a structural member.

3.1.2.1 Recycled materials. All materials, equipment and articles incorporated in the work covered by this specification shall be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means material which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that use of used or rebuilt products is allowed under this specification.

3.1.2.2 Protective treatment. When materials that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage are used, they shall be protected against such deterioration in such a manner that shall in no way prevent compliance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

3.1.2.3 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.2.4 Bonding. All metal parts shall be specifically bonded together to prevent the possibility of sparking.

3.1.3 Design and construction. The crane shall be constructed so that parts do not work loose in service and shall withstand the loads, shock, vibrations, and other conditions incident to operation,

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shipping, and storage with minimum loss of time for maintenance, repair, and service. The design of the crane shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, removal, or replacement. The crane shall comply with all requirements of MIL-S-8512 and MIL-STD-1472.

3.1.3.1 Reliability. The crane, including all components, shall have a Mean-Time-Between-Failures (MTBF) of not less than 300 hours demonstrated, in accordance with Test Plan XVIIC of MIL-STD-781. A failure is defined as the event or inoperable state in which any item or part of an item does not, or would not, perform as previously specified. Definitions of reliability terms not included herein shall be as found in MIL-STD-721 and MIL-STD-781. In the event of a conflict between these documents, MIL-STD-781 shall take precedence. The crane shall be designed for a life expectancy of at least 20 years.

3.1.3.2 Maintainability. The crane shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the crane. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing, and replacing parts and components.

d. Use or readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime of no greater than one hour.

g. Maximum corrective maintenance downtime of no greater than four hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 10 minutes.

3.1.3.3 Foolproofness. Where improper installation of an item could cause malfunction, an unsymmetric mounting shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins or lock washers shall not be used for this purpose.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be located so they are readily accessible and do not require special tools for proper servicing.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings conforming to SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.3.6.1 Hydraulic fluid. Hydraulic fluid shall be in accordance with MIL-H-83282.

3.1.3.7 Protective features. All space in which work is performed during operation, service and maintenance of the unit shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

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3.1.3.8 Component protection. All oil, hydraulic and air system tubing and all electrical lines shall be securely clipped to the frame or the body structure and, except where a through frame connector is necessary, shall be furnished with metal protective looms or grommets at each point where they pass through a panel or structural member.

3.1.4 Environmental conditions. The crane shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full time exposure to temperatures ranging from -40°F to +125°F for operation; -60°F to +160°F for storage.

b. Full time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0mm to 4.5mm, falling at a rate of 5.2 inches per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0ml/80cm^2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1,750ft/min, concentration 0.5g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5,700ft/min, concentration 0.0773g/ft^3.

g. Part-time (15 percent of the time) exposure to vibration as described in Table 514.3-I, Category 1, of MIL-STD-810.

3.1.5 Weight distribution. The weight distribution of the unit shall not be unbalanced or improperly distributed.

3.1.6 Electromagnetic interference suppression. The crane shall comply with the requirements of MIL-STD-461 for electromagnetic interference suppression for this type equipment.

3.1.7 Air transportability. The vehicle shall be air transportable without shoring or disassembly of any components in accordance with MIL-STD-1791 in Types C-130, C-141, and C-5 Aircraft and shall be air transportable in a Type C-17 Aircraft. The crane shall be capable of being towed onto/off of the aircraft. Preliminary design guidelines for the Type C-17 are 142 inches maximum height, 204 inches maximum width, 812 inches maximum length, 9° ramp angle, and floor load capacity equal to that allowed for the Type C-5 Aircraft.

3.1.7.1 Tiedown clearance. Sufficient clearance shall be provided around all attachment points to allow for the attachment of cables and chains without interference from any other component of the crane. Each tiedown point shall have a minimum clear opening of four inches to allow attachment of chain tiedown with a hook and shall be in accordance with MS21236-2.

3.1.8 Factor of safety. Except as otherwise specified, all load bearing components shall be designed with a factor of safety of three, based on the minimum material yield strength.

3.2 Performance.

3.2.1 Capacity.

3.2.1.1 Rated capacity. The crane shall be capable of a 1,500 pound minimum lift through any position of the boom swing. The crane operational requirements are as follows:

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a. Vertical lift height of 27 feet (measured from the lifting hook to the ground.)

b. Horizontal reach of 15 feet (measured from the lifting hook to the center of mast rotation.)

c. Vertical angle of boom articulation is from 0° to 180°.

d. Mast swing is 360° continuous rotation.

3.2.1.2 Proof load. The crane shall be capable of lifting a 2,250 pound load to a hook height of either 27 feet or at a reach of 15 feet from the crane mast horizontal with no damage, permanent deformation or instability of the crane.

3.2.2 Boom swing. The crane shall be capable of swinging the rated capacity through a 360° continuous rotation. This can be done manually or hydraulically.

3.2.3 Operating speed.

3.2.3.1 Lift speed. The lift speed with rated power supply shall not be less than five feet per minute with a load of 1,500 pounds and shall not exceed 10 feet per minute with no load. With input voltage 10 percent greater than nominal, the lifting speed of the crane with no load shall not exceed 12 feet per minute. With input voltage 10 percent less than nominal, the crane shall be able to start and lift any load up to 1,500 pounds at a speed of not less than 3.3 feet per minute.

3.2.3.2 Lowering speed. With input power supply as specified in 3.7.1, the lowering speed to the crane in any load condition shall not be less than 3.3 feet per minute and shall not exceed 10 feet per minute.

3.2.4 Hoist brake performance. The winch/hoist mechanism brake shall be capable of stopping a 1,500 pound load within a distance of one inch while lifting and within a distance of 1-1/2 inches while lowering. Winch/hoist mechanism brake shall continue to hold during a power failure.

3.2.5 Mobility. The crane shall be collapsible without disassembly and towable on itself. The collapsed crane assembly shall comply with the mobility requirements of MIL-M-8090, Type II, Group A.

3.2.6 Operation on slopes. The crane shall be capable of operation on slopes of up to three percent.

3.2.7 Functional Requirements. The crane must be capable of being set-up, operated to complete a task and then returned to the transport mode while remaining in the same location. The crane must be capable of performing the following operations without interfering with the aircraft:

Functional Analysis

The F-16 aircraft wings, vertical stabilizer, canopy, seat(s), ammunition drum and leading edge control and drive unit are removable components, which could require removal and installation during Organizational and Intermediate level maintenance. Parameters of these components are as follows:

1. Wings, P/N 16W030, WUC's 11LO and 11MO

a. Weight - 650 pounds (Leading Edge Flap and Flaperon removed).

b. Size - Wing Root Length 12.5 feet
Tip Length 3.4 feet
Chord Length 12.3 feet

c. Location on Aircraft - 72 inches above static ground.

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2. Vertical Stabilizer, P/N 16T7200, WUC 11JO

a. Weight - 225 pounds

b. Size - Base Length 100 inches
Tip Length 47.5 inches
Vertical Height 101 inches

c. Location on Aircraft - Base 96 inches above static ground.

- Tip 198 inches above static ground.

3. Leading Edge Flap Drive and Control Unit, P/N 16M002-3, WUC 14DAO

a. Weight - 44 pounds

b. Size - Length 10 inches (fore and aft)
Width 16 inches
Height 13.5 inches

c. Location on Aircraft - Access from top of fuselage at BL 0.00, FS 284-294, WL 125 (105 inches above static ground).

4. Canopy Assembly, F-16A P/N 16K0510, WUC 12CA0-A
F-16B P/N 16K2510, WUC 12CA0-B

Characteristics of the F-16A/B canopy include the following:

a. One piece unit, hinged at the rear extremity, opening upward to a maximum actuated travel of 35 degrees (F-16A) and 31 degrees (F-16B). The opening travel to the point where the canopy disengages from its hinge is 49 degrees (F-16A) and 41 degrees (F-16B).

b. Physical description:

		F-16A	F-16B
1.	Weight1	90 pounds	300 pounds
2.	Size	91 inches	152 inches
	Width	30 inches	30 inches
	Height	24 inches	28 inches

c. Location on Aircraft - Front end of base 88 inches above static ground line, sloping up and aft.

d. Lifting Points - Integral to canopy structure.

e. Transparency - Coated polycarbonate plastic.

5. Ammunition Drum, P/N 132D1869, WUC 75ABC

a. Physical Description:

1. Weight - 140 pounds (empty), 375 pounds (loaded with 20 MM ammo) (approximately)

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2. Diameter - 22.5 inches

3. Length - 24.0 inches

b. Location on aircraft - access from top of fuselage FS 254-65, BL 0.00, WL 106.5.

c. Lifting provisions - Two plates bolted to the drum flange and having a rectangular slot compatible with the 5/8-inch lug beam assembly hook, P/N T1-F66.

6. Seat, F-16A P/N J114716, WUC 12E00-A
F-16B P/N J114716, WUC 12E00-B

The seat has the following leading characteristics:

a. Physical description:

1. Weight - 150 pounds

2. Length - 29 inches

3. Width - 20 inches

4. Height - 52 inches

b. Location on aircraft - Access from top of fuselage with canopy removed. F-16A and F-16B front seat is at FS160 (top of rail) and F-16B rear seat is at FS230 (top of rail). (Rails are 34.5 degree angle to aircraft vertical.)

Unscheduled maintenance can require the removal and installation of the above components. A hoisting device will be required to accomplish this removal and installation when overhead hoisting shop facilities are not available, with external fuel tanks installed. Aircraft may be located in a Harden Shelter.

3.3 Frame Assembly.

3.3.1 Towbar. The frame assembly shall be equipped with a towbar designed for towing by vehicles having an adjustable (without tools) pintle hook. The height shall be adjustable from 10 to 30 inches. The towbar shall incorporate a lunette eye conforming to MS 51336.

3.3.2 Landing gear. The frame assembly shall be equipped with an adjustable retractable landing gear. The landing gear shall remain positively locked when placed in either the raised or lowered position. The landing gear shall be designed to support, raise and lower the crane when not attached to a towing vehicle and shall be equipped with a caster of sufficient capacity to withstand loads imposed.

3.3.3 Wheels, tires and tubes. The crane shall be equipped with commercial type wheels and tires which meet the Tire and Rim Association requirements for this application. When equipped with tube type tires, inner tubes and tire flaps shall be provided in accordance with the Tire and Rim Association recommendations.

3.3.4 Outriggers. The crane shall be equipped with sufficient outriggers to comply with the stability requirements of 3.2.1.2. A lockout mechanism shall be incorporated to prevent the crane from being placed in an unsafe mode.

3.4 Operator station.

3.4.1 Controls. All controls and indicators shall be located within easy reach of the operator while standing in front of the operator station. All controls shall provide for smooth operation of the crane.

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3.5 Winch, mast and boom.

3.5.1 Winch/hoisting mechanism. The hoist mechanism shall be hydraulic motor driven and shall incorporate an automatic brake capable of holding the maximum rated load. The system shall provide for not less than 16 feet of cable take up from boom tip.

3.5.1.1 Wire rope. The load hoist rope shall be 6X37 improved plow steel, independent wire rope center and have a strength factor of not less than 3.5 in accordance with SAE J959. Sufficient rope shall be provided for all conditions of operation and erection of the crane specified herein.

3.5.1.2 Hook. The hook shall be of sufficient weight to insure payout of the cable with no load on it. The hook shall be of the swivel type with a safety latch. The hook shall have a minimum factor of safety of four based on the ultimate strength.

3.5.1.3 Anti-two block system. An anti-two block system shall be provided to prevent capable breakage and wedging or binding of the hook when at boom end.

3.5.2 Mast. The crane mast shall lock securely in an upright position for operation and shall fold or collapse for storage and air transportability. A hydraulic erection system shall be provided. The crane shall be erected utilizing its own power.

3.5.3 Boom. The boom shall be of the hydraulically operated articulated type. It shall consist of two pivoting sections connected to the stationary mast. The boom shall be capable of lifting from 0° to 180° with respect to the vertical of the mast.

3.6 Hydraulic system. The hydraulic system shall include the pump, cylinders, reservoir, filter, pressure relief valve and all necessary lines, restrictors and control valves to insure positive control of the winch/hoisting mechanism and boom in all operations and in the event of hydraulic pressure loss. The pump shall have sufficient capacity to energize all hydraulic powered components operating simultaneously. The crane shall be equipped with a hand pump of sufficient capacity to operate all functions of the crane. Provision shall be made for bleeding all air from the system. The manual operation shall be in accordance with MIL-STD-1472. All hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure and a proof pressure of two times the maximum operating pressure.

3.7 Electrical system.

3.7.1 Power supply.

3.7.1.1 External power supply. The crane shall be designed for operation on a nominal input power supply of 22 volt AC 50/60 Hz single phase and 28 volt DC. The crane shall be capable of operation with an input power supply voltage 10 percent greater than or less than nominal.

3.7.1.2 Internal power supply. The crane shall be equipped with two heavy duty 12 volt DC batteries equivalent to MS 35000-4 (100 AMP-HR). This system shall provide 24 volt DC for emergency operation. A battery charging system shall be supplied operating on 110 volt AC 50/60 Hz. The crane shall have an emergency hand pump for operation in the event of a total power failure. The operator's effort to lift the rated load shall be in accordance with MIL-STD-1472.

3.7.2 Receptacles. A receptacle conforming to MS 25488 shall be provided for connection to a 28 volt DC power supply. A receptacle shall be provided for connection to a 220 volt 50/60 Hz single phase power supply. A cover shall be provided for each receptacle, securely held in the closed position by a spring. The receptacles shall be located in a protected location which is easily accessible for connection of a power cable.

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3.7.3 Power cables. The crane shall be furnished with a power cable and plug of not less than thirty feet in length for connection of the 220 volt 50/60 Hz single phase power supply. An additional power cable shall be furnished in accordance with MS 90347-8.

3.7.4 Overload protection. All circuits shall be protected by circuit breakers in lieu of replaceable fuses.

3.8 Finish and protective coatings.

3.8.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808 and/or MIL-T-704, except that commercial (see 6.3.2) components may be prepared in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish paint.

3.8.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defect and shall be capable of passing the test described in paragraph 4.8.10 of MIL-C-83286.

3.8.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the crane and the interior surfaces of compartments shall be painted with polyurethane paint, Forest Green, Color Number 24052, of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be DuPont Imron, or equal. All exterior trim items shall be painted with the same paint as the exterior of the crane.

NOTE

"Equal" is defined as having performance identical to or superior to DuPont Imron when subjected to the following tests:

a. Tests in accordance with MIL-C-46168:

(1) Paragraph – 4.3.17 – Flexibility.

(2) Paragraph – 4.3.23 – Accelerated weathering.

b. Tests in accordance with MIL-C-83286:

(1) Paragraph – 4.8.6 – Low temperature resistance.

(2) Paragraph – 4.8.10 – Resistance to taping.

(3) Paragraph – 4.8.12 – Pencil hardness.

3.8.2.2 Desert Sand. When specified (see 6.2), the outer surface of the crane and the interior surfaces of compartments shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313, of FED-STD-595. All exterior trim items shall be painted with the same paint as the exterior of the crane.

3.8.3 Rustproofing. Unless otherwise specified (see 6.2), the crane shall be rustproofed in accordance with the contractor's standard commercial practice.

3.9 Markings. Unless otherwise specified, markings for cranes painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.9.1 Identification markings. Unless otherwise specified (see 6.2), the crane shall be marked for the appropriate service in accordance with MIL-STD-1223.

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3.9.2 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.9.3 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the crane in a readily accessible location. The identification shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURER

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.9.4 Control markings. All controls, valves, gauges, and indicators used in the operation of the crane shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, etc, will neither fade nor eradicate them.

3.9.5 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed on the operator control panel. The instructions shall be made clear, concise and adequate to enable operation of the crane without damage to the equipment or injury to personnel. Instruction panels shall be made from sheet aluminum of not less than 0.050 inch thickness, anodized or etched to produce raised black or dark contrasting markings with a light background and a border of at least 0.25 inch width.

3.9.6 Loading capacity chart. A permanent chart indicating the safe load shall be mounted in clear view of the operator.

3.9.7 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperature.

3.10 Dimensions. The overall dimensions of the crane shall not exceed the following dimensions while in the collapsed configuration.

a. Length – 250 inches.

b. Width – 96 inches.

c. Height – 102 inches.

d. Weight – 5,500 pounds. (Maximum)

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3.11 Workmanship. The crane, complete with all parts and accessories, shall be fabricated in a thoroughly workmanship manner. Particular attention shall be given to freedom from blemishes, burrs, defects, sharp edges, accuracy of dimensions, radii of fillets, marking of parts and assemblies, thoroughness of welding, brazing, soldering, riveting and painting, alignment of parts, tightness of assembly fasteners, etc. The crane shall be thoroughly cleaned of all foreign matter.

3.11.1 Welders and welding.

3.11.1.1 Welders. All welders employed in the fabrication of the crane shall be certified in accordance with the application requirements of MIL-STD-1595, the American Welding Society or the American Society of Mechanical Engineers.

3.11.1.2 Welding. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.11.3 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.11.4 Riveting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.11.5 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities, Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.11.6 Cleaning. The crane shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust metal particles, pipe compound, and other foreign matter shall be removed after final assembly prior to painting (see 3.8.1).

3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in accordance with MIL-STD-130. (Note that MIL-STD-130 does not require marking of commonly known commercial parts which present no identification problem or parts in assemblies which are not normally subject to disassembly or repair.)

3.13 First Article. The contractor shall furnish one crane for first article tests (see 4.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services *conform* to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system

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or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the crane shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to insure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data shall be recorded at intervals not to exceed 30 minutes, where applicable:

a. Crane load (pounds).

b. Input voltage.

c. Input current (amperes).

d. Ambient temperature (degrees Fahrenheit).

4.3.3 Power supply. Each test, except 4.6.3 which will be at 220 volts 50/60 Hz single phase, shall be performed using each power supply specified in 3.7.1.

4.4 First article testing.

4.4.1 First article test sample. The first article test sample may be a preproduction item or a production item that meets the requirements of this specification. The sample shall be identical to the items procured under the terms of this specification. Approval of a first article sample shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article test. The first article tests shall consist of all tests listed under 4.6, performed in order.

4.5 Acceptance testing. Each crane shall be subjected to the examination of product as described in 4.6.1. Failure to comply with any portion of this specification or inability to perform any crane function shall be cause for rejection.

4.6 Test methods.

4.6.1 Examination of product. The crane shall be inspected to determine compliance with the requirements of this specification. A thorough operational check shall be made to insure the proper functioning of the mechanical parts of the crane, accessibility of all controls and levers, interchangeability of removable parts, installation of equipment and safety of operation. Any certifications necessary to document compliance with all requirements shall be supplied at the first article test and shall be included in the first article test and shall be included in the first article test report when specified (see 6.5).

4.6.2 Electromagnetic interference measurement. The crane shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.1.6.

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4.6.3 Proof load test.

4.6.3.1 Static test. A load of 3,000 pounds shall be attached to the cable hook and supported by the crane for a period of not less than 30 minutes. The load shall be removed and the crane shall be inspected. Evidence of cracks, permanent distortion, creep or any other failure shall be cause for rejection.

4.6.3.2 Dynamic test. A load of 2,250 pounds shall be attached and lifted to a hook height of 27 feet. The load shall be lowered and the crane shall be inspected. Evidence of cracks, permanent distortion, creep or any other failure shall be cause for rejection.

4.6.4 Load swing test. A load of 1,500 pounds shall be attached to the cable hook and lifted to a height of two feet. The load shall be swung through 360° of travel and timed for two revolutions in each direction. The values shall comply with 3.2.2. The force required for swing in each direction shall be measured and recorded to demonstrate compliance with MIL-STD-1472.

4.6.5 Manual operation test. A load of 1,500 pounds shall be attached to the cable hook. The load shall be lifted using the hand pump to the maximum height and lowered for 10 cycles. The force on the hand pump necessary for operation in each direction shall be measured and recorded to demonstrate compliance with MIL-STD-1472.

4.6.6 Operating speed. The crane shall be tested to demonstrate compliance with 3.2.3.1 and 3.2.3.2 for each specified load condition and input power supply.

4.6.7 Power requirements. A load of 1,500 pounds shall be attached to the cable hook. The load shall be lifted to the maximum height at maximum winch speed and lowered using each input power supply. The maximum power required for each input power supply shall be measured and recorded to demonstrate compliance with 3.7.1.

4.6.8 Mobility tests.

4.6.8.1 Unloaded. The crane shall be collapsed and towed for a distance of at least 10 miles over smooth concrete or macadam paving at an average speed of not less than 20 miles per hour. Any evidence of instability or failure to comply with MIL-M-8090 shall be cause for rejection.

4.6.9 Environmental tests. The crane shall be subjected to the following environmental tests. An engineering evaluation may be performed in lieu of the testing specified in 4.6.9.3, 4.6.9.4 and 4.6.9.5. The engineering evaluation shall consist of a written detailed analysis of the crane and all its parts and components to verify compliance with 3.1.4 b, c, and d.

4.6.9.1 High temperature test. The collapsed crane shall be placed in an environmental chamber and soaked for a minimum of 24 hours at 125°F, plus or minus 2.5°F. At the conclusion of the soak while still in the chamber, the crane shall be erected and operation of all crane functions shall be demonstrated using each power supply. The crane shall then be collapsed. Evidence of overheating of any component or inability to perform any function shall be cause for rejection.

4.6.9.2 Low temperature test. The collapsed crane shall be placed in an environmental chamber and soaked a minimum of 24 hours of -40°F, plus or minus 2.5°F. While still in the chamber, the crane shall be erected and operation of all crane functions shall be demonstrated using each power supply. The crane shall then be collapsed. Inability to perform any function shall be cause for rejection.

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4.6.9.3 Humidity. The crane shall be subjected to humidity in accordance with MIL-STD-810, Method 507, Procedure I.

4.6.9.4 Salt Fog. The crane shall be subjected to salt fog in accordance with MIL-STD-810, Method 509, Procedure I.

4.6.9.5 Dust. The crane shall be subjected to dust in accordance with MIL-STD-810, Method 510, Procedure I.

4.6.10 Endurance Test. A load of 1,500 pounds minimum shall be attached to the cable hook. The load shall be lifted to the maximum hook height using the boom pivot and the cable take up mechanism. The load shall then be stopped and rested for 15 seconds maximum and lowered to ground level for 100 cycles. A one minute rest shall be permitted after every fifth cycle. During the one minute rest, the hydraulic fluid temperature shall be recorded. At the conclusion of the endurance test, the crane shall be inspected and any damage recorded.

4.6.11 Hoisting brake tests.

4.6.11.1 Hoisting brake endurance test. A load of 1,500 pounds shall be attached to the cable hook. The hoist shall be brought to uniform speed in the upward direction, stopped and rested 15 seconds maximum, brought to uniform speed in the downward direction, stopped and rested for 15 seconds maximum for 200 cycles. A rest period of 10 minutes shall be permitted after each 50 cycles.

4.6.11.2 Winch brake performance test. The cable brake shall be tested to demonstrate conformance with the requirements of 3.3.4.

4.6.12 Air transportability.

4.6.12.1 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.7. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard Types C-130, C-141, C-17, and C-5 Aircraft.

4.6.12.2 Preparation for air transport and operation. Compliance with the requirements of 3.1.7 for preparation of the crane for air transport and for operation shall be demonstrated. Actual times required and a list of hand tools used shall be recorded in the test report.

4.6.13 Structural analysis. A structural analysis shall be performed on the load supporting components of the crane assembly. Maximum stress shall not exceed one third the material minimum yield strength. A Goodman analysis shall be performed on all load supporting fasteners. With maximum installation torque, this analysis shall indicate infinite life (5×10^6 cycles for nonferrous fasteners) for a 0 to 1,500 pound to a 0 load cycle.

4.6.14 Slope operation. The crane shall demonstrate compliance with 3.2.6 with a 1,500 pound load.

4.6.15 Final inspection. The crane shall be completely disassembled for inspection of all components. Evidence of cracks, permanent distortion, undue wear, leaks or any other failure shall be cause for rejection.

4.6.16 Functional analysis. A detailed functional analysis shall be submitted in order to show that each requirement in paragraph 3.2.7 is met. Failure to accomplish any task shall be cause for rejection of unit.

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4.7 Refurbishment. After approval of the first article test report when specified (see 6.5), the first article unit shall be restored to like new condition.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 1,500 pound crane covered by this specification is intended for support equipment for F-16 aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.8.2.1, 3.8.2.2).

c. Markings required (see 3.9).

d. If any other than the contractor's standard commercial rustproofing is required (see 3.8.4).

e. Identification markings required if other than in accordance with MIL-STD-1223 (see 3.9.1).

6.2.1 First article. When a first article inspection is required, the items should be first article samples. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Definitions.

6.3.1 Commercial. A manufacturer's standard (product) having been marketed, in production, and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. A item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes, or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items for foreign military sales.

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6.4 Data requirements. The following Data Item Descriptions (DID's) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
	DI-A-7088	Conference Agenda	See DD Form 1423
	DI-A-7089	Conference Minutes	See DD Form 1423
	DI-A-4901	Test Procedures	See DD Form 1423
6.5	DI-A-4902	Test Report	See DD Form 1423
4.4.3	DI-T-23731A/T	Notification of Tests	See DD Form 1423
4.6.13	UDI-A-23122A/T	Shock Dynamic Design Analysis Report	See DD Form 1423
	DI-A-3011	Still Photo Coverage	See DD Form 1423

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-1, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.5 Test report. Upon the completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operation identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

g List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

h. Test report for each component covered by a document listed in the DODISS.

i. Plan for testing production units.

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6.6 Subject term (key word) listing.

Base
Boom
Casters
Collapsible
Hook
Hydraulic
Lunette eye
Mast
Portable
Wire rope
220 volt AC
28 volt DC

Custodians:
Air Force-99

Preparing activity:
Air Force-84

(Project 3950-F018)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-83810 (84)	4 Oct 91

3. DOCUMENT TITLE
CRANE, FLOOR, PORTABLE, 1,5000 POUND CAPACITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If Applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
David Owen	912-926-2074	932-2074
c. ADDRESS (Include Zip Code) WR-ALC/LZMD Robins AFB GA 31098-5609	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete.

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